

C4 past final worksheet

1. (Fall 2024, #10) Consider the function F defined on the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ by:

$$F(x) = \int_0^x \frac{1}{t^2 + 1} dt$$

You must carefully justify your answers and reasoning.

- (a) Show that $F(x)$ has an inverse on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then determine the value of $(F^{-1})'(0)$.

- (b) Let $G(x) = F(\tan(x))$. Calculate the derivative of $G(x)$. Show that $G(x) = x$ on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

- (c) Determine the value of $\int_{-1}^1 \frac{1}{t^2 + 1} dt$.

2. (Spring 2024, #9) Consider the following integral function:

$$I(x) = \int_e^{e^x} \frac{1}{t^3 + 3} dt$$

- (a) Find $I'(x)$.

- (b) Prove that $I(x)$ is an invertible function.

- (c) Find $(I^{-1})'(0)$.

3. (Fall 2023, #9) Given

$$f(x) = 2x + \ln(x^2 - 3)$$

(a) Show that f is one to one on the domain $(\sqrt{3}, +\infty)$.

(b) Find $f(2)$.

(c) Compute $(f^{-1})'(4)$.

(d) Compute $(f^{-1})'(f(3))$.

4. (Spring 2023, #10) The function

$$P(h) = 50 + 13.3 \int_{65}^h e^{-\frac{1}{2}\left(\frac{x-65}{3}\right)^2} dx$$

gives the percent, P , of people in the world that are shorter than h inches tall. For example, a calculator can show that $P(68) = 84$, indicating that 84% of people in the world are shorter than 68 inches tall.

- (a) Show that $P(h)$ has an inverse function. You must use calculus to answer this question, i.e., you cannot just reason from the given context.

- (b) If we let $h(P)$ be the inverse of $P(h)$, find $h'(84)$.

- (c) Which of the following are correct interpretations of the quantity $h'(84)$? Circle all that apply.
- A. The slope of the line tangent to $P(h)$ at $(68, 84)$.
 - B. The rate of change of P with respect to h at $(68, 84)$.
 - C. The rate of change of h with respect to P at $(84, 68)$.
 - D. The approximate share of people in the world whose heights are between 68 and 69 inches.
 - E. The approximate difference in height between someone who is taller than 84% of people in the world and someone who is taller than 85% of people in the world.

5. (Spring 2022, #9) Define $F(x) = \int_1^{x^3} \frac{1}{1+t} dt$ for $x > -1$.

(a) Compute $F'(x)$.

(b) Show that F is one-to-one.

(c) Compute $(F^{-1})'(0)$.